Day 08 – Cloud Server Setup:Nginx & Web Deployment

Part 1 – Launch Cloud Instance & SSH Access

Step 1: Login to AWS

Step 2: Launch EC2 Instance

Click:

`Launch Instance`

Configure:

Setting	Value
Name	day08-nginx-server
AMI	Ubuntu Server 22.04 LTS
Instance Type	t2.micro (Free Tier)
Key Pair	Create new key pair
Security Group	Allow SSH + HTTP

Step 3: Create Key Pair

Choose:

- Key Pair Name: `day08-key`
- Type: `.pem`

Download the `.pem` file carefully.

Example:

```
day08-key.pem
```

Store it safely in Downloads folder.

Step 4: Configure Security Group

Enable these:

Type	Port
SSH	22
HTTP	80

This allows:

- SSH access
- Website access from browser

Click:

```
Launch Instance
```

Step 5: Copy Public IP

After instance launches:

Copy:

`Public IPv4 address`

Example:

```
13.233.xx.xx
```

Step 6: Connect via SSH

Windows (Git Bash / PowerShell)

Move to Downloads folder:

```
cd ~/Downloads
```

Give key permission:

```
chmod 400 day08-key.pem
```

Connect to server:

```
ssh -i day08-key.pem ubuntu@YOUR_PUBLIC_IP
```

Example:

```
ssh -i day08-key.pem ubuntu@13.233.xx.xx
```

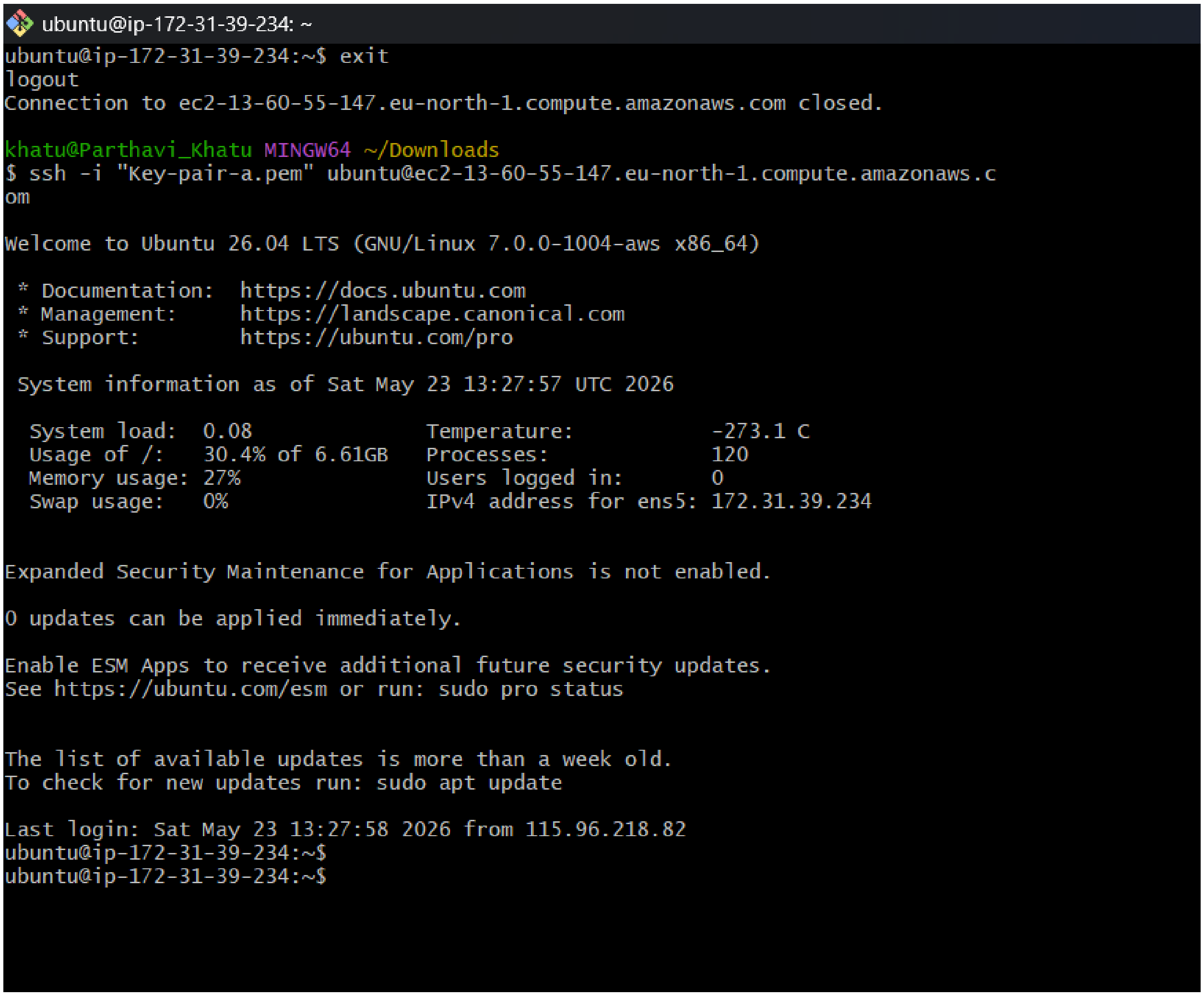
Expected Output

You should see something like:

```
Welcome to Ubuntu 22.04 LTS
ubuntu@ip-172-31-x-x:~$
```

☐ 📷 Take Screenshot:

- SSH terminal connected to EC2



Part 2 – Install Docker & Nginx

Step 1: Update System

Run:

```
☐sudo apt update && sudo apt upgrade -y
```

Step 2: Install Docker

Install Docker:

```
sudo apt install docker.io -y
```

Start Docker:

```
sudo systemctl start docker
```

Enable Docker on boot:

```
sudo systemctl enable docker
```

Check status:

```
sudo systemctl status docker
```

Expected:

```
active (running)
```

Step 3: Install Nginx

Install Nginx:

```
sudo apt install nginx -y
```

Start Nginx:

```
sudo systemctl start nginx
```

Enable Nginx:

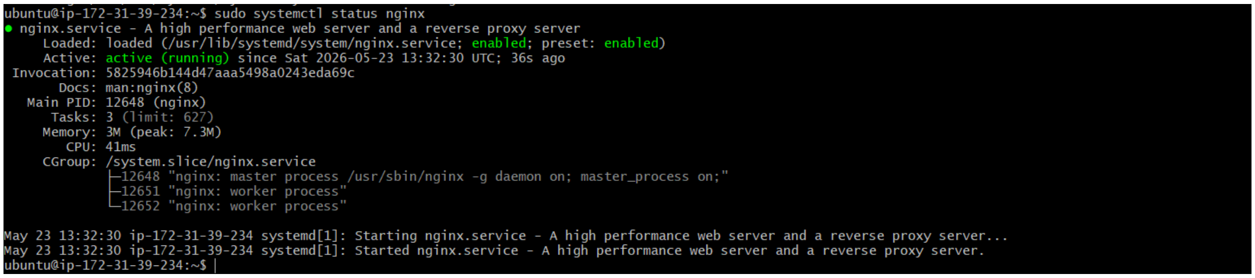
```
sudo systemctl enable nginx
```

Check status:

```
sudo systemctl status nginx
```

Expected:

```
active (running)
```



```
ubuntu@ip-172-31-39-234:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-05-23 13:32:30 UTC; 36s ago
 Invocation: 5825946b144d47aaa5498a0243eda69c
       Docs: man:nginx(8)
   Main PID: 12648 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 41ms
     CGroup: /system.slice/nginx.service
             ├─12648 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─12651 "nginx: worker process"
             └─12652 "nginx: worker process"

May 23 13:32:30 ip-172-31-39-234 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 23 13:32:30 ip-172-31-39-234 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-39-234:~$ |
```

Part 3 – Verify Web Access

Open browser:

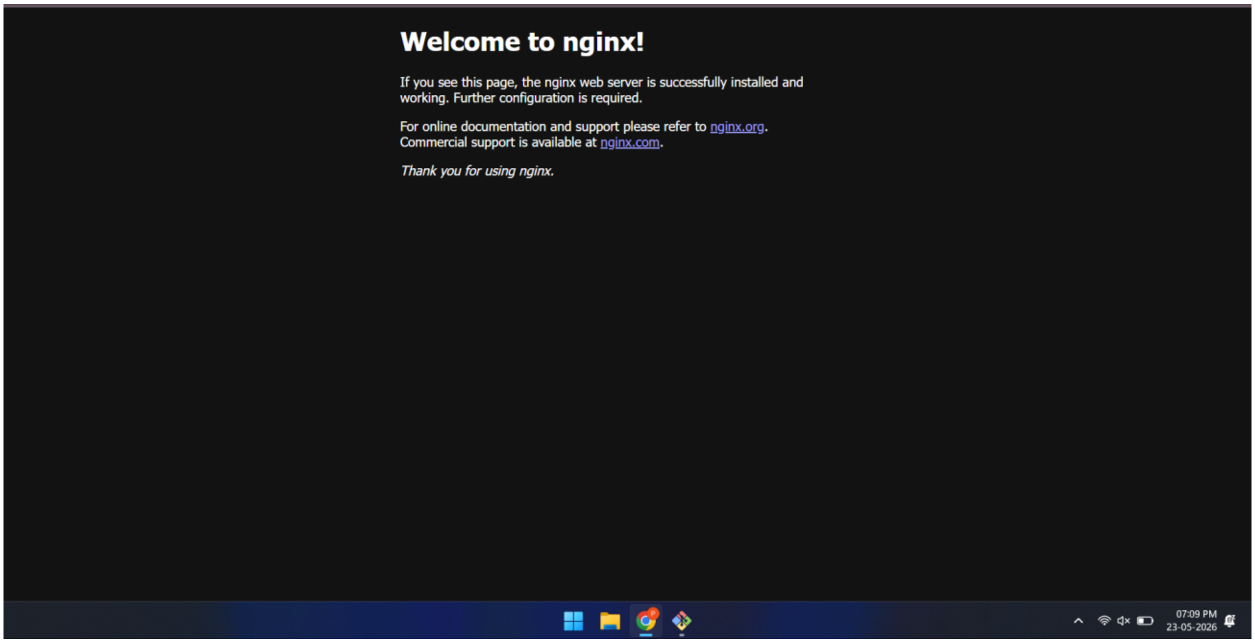
```
http://YOUR_PUBLIC_IP
```

Example:

```
http://13.233.xx.xx
```

Expected Result

You should see:



“Welcome to nginx!”

📸 Take Screenshot:

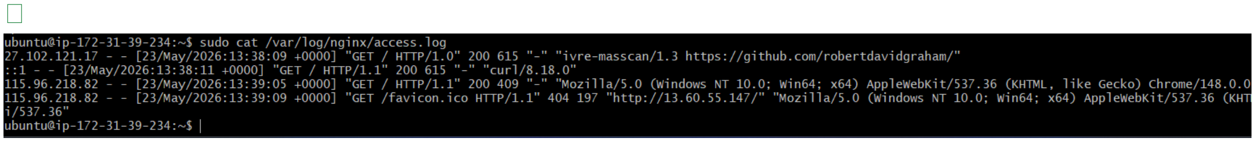
- Nginx welcome page in browser

Part 4 – Extract Nginx Logs

Step 1: View Nginx Logs

Access log:

```
sudo cat /var/log/nginx/access.log
```



```
ubuntu@ip-172-31-39-234:~$ sudo cat /var/log/nginx/access.log
27.102.121.17 - - [23/May/2026:13:38:09 +0000] "GET / HTTP/1.0" 200 615 "-" "ivre-masscan/1.3 https://github.com/robertdavidgraham/"
::1 - - [23/May/2026:13:38:11 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.18.0"
115.96.218.82 - - [23/May/2026:13:39:05 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0
115.96.218.82 - - [23/May/2026:13:39:09 +0000] "GET /favicon.ico HTTP/1.1" 404 197 "http://13.60.55.147/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHT
i/537.36"
ubuntu@ip-172-31-39-234:~$ |
```

Error log:

```
sudo cat /var/log/nginx/error.log
```

You will see browser request entries.

Step 2: Save Logs to File

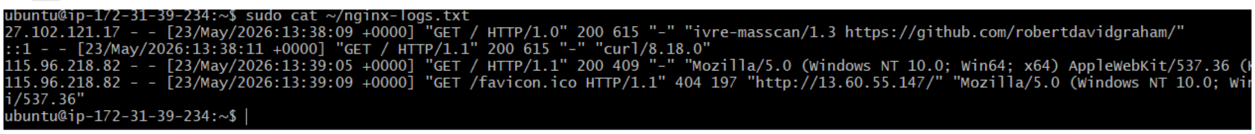
Copy access logs:

```
sudo cp /var/log/nginx/access.log ~/nginx-logs.txt
```

Verify:

```
cat ~/nginx-logs.txt
```

📷 Take Screenshot:



- Terminal showing nginx-logs.txt contents

Step 3: Download Log File to Local Machine

Open NEW terminal on your local machine.

Move to Downloads:

```
cd ~/Downloads
```

Run:

```
scp -i day08-key.pem ubuntu@YOUR_PUBLIC_IP:~/nginx-logs.txt .
```

Example:

```
scp -i day08-key.pem ubuntu@13.233.xx.xx:~/nginx-logs.txt .
```

Check downloaded file:

```
ls
```

You should see:

```
nginx-logs.txt
```

```
Nginx-logs.txt---------------
ubuntu@ip-172-31-39-234:~$ sudo cat nginx-logs.txt
27.102.121.17 - - [23/May/2026:13:38:09 +0000] "GET / HTTP/1.0" 200
615 "-" "ivre-masscan/1.3 https://github.com/robertdavidgraham/"
::1 - - [23/May/2026:13:38:11 +0000] "GET / HTTP/1.1" 200 615 "-"
"curl/8.18.0"
115.96.218.82 - - [23/May/2026:13:39:05 +0000] "GET / HTTP/1.1" 200
409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
115.96.218.82 - - [23/May/2026:13:39:09 +0000] "GET /favicon.ico
HTTP/1.1" 404 197 "http://13.60.55.147/" "Mozilla/5.0 (Windows NT
10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/148.0.0.0 Safari/537.36"
```